

Key Insights

Finding, Evidence, Business Meaning, Recommended Action

Each insight follows: **Finding** → **Evidence** → **Business Meaning** → **Recommended Investigation/Action**. All figures are pulled directly from `data/processed/monthly_revenue_processed.csv` — none are estimated.

1. Early-period growth percentages are real but not representative

Finding: The two largest MoM growth percentages in the entire series occur in the first two months of data.

Evidence: Nov-2016 +144.19% (43,000 → 105,000); Dec-2016 +140.95% (105,000 → 253,000).

Business Meaning: These are mathematically correct, but they reflect a near-zero starting base, not a repeatable growth rate. If reported without context (e.g., in a single "average MoM growth" headline number), they would materially overstate the business's typical growth rate.

Recommended Action: Exclude or footnote the first 2-3 months whenever reporting an average/typical MoM growth rate. Use median MoM growth, or a "from month 4 onward" filter, alongside the full series.

2. Revenue shows a clear ramp-then-plateau shape, not sustained compounding

Finding: Revenue climbs from 43,000 to a period peak of 1,061,000 (May-2018), but the growth rate visibly slows after the Nov-2017 spike.

Evidence: 3-month rolling average moves from 133,667 (Dec-2016) to 640,667-674,000 (mid-to-late 2017), and settles in a 972,667-1,045,333 band from Apr-2018 through Aug-2018 (`data/processed/monthly_revenue_processed.csv`, `rolling_3m_avg` column).

Business Meaning: The business has moved from a launch-growth phase into a maturity phase relative to its current base. This is a normal pattern for a young business, not necessarily a warning sign on its own.

Recommended Action: Investigate what specifically changed after Nov-2017 (a new marketing channel, category expansion, referral program, etc.) using data outside this dataset's scope — order-level, marketing spend, or channel data would be needed to say more.

3. The Nov-2017 spike is a genuine outlier, followed by reversion — not a new baseline

Finding: Nov-2017 revenue (1,027,013) is far above the surrounding months (Oct-2017: 635,000; Dec-2017: 780,000), and Dec-2017 shows the worst MoM growth in the series (-24.05%).

Evidence: `mom_growth_pct` for Nov-2017 = +61.73%, for Dec-2017 = -24.05% — the single largest month-over-month swing pair in the dataset outside the initial ramp.

Business Meaning: A single unusually strong month followed by a sharp pullback is consistent with a one-off event rather than a permanent step-change in the business's run rate. This dataset cannot say what the event was (or whether it recurs) - it has no calendar, promotion, or order-level data. Jan-2018 (952,000) confirms this — it settles well below the Nov-2017 peak but above the pre-spike baseline.

Recommended Action: Identify what happened in Nov-2017 specifically (this dataset alone cannot say) and evaluate whether it is a repeatable, plannable event (e.g., an annual seasonal calendar) or a non-repeatable one-off.

4. There is no evidence of a sustained decline — the most recent dip is isolated

Finding: The longest run of consecutive MoM-negative months anywhere in the 23-month series is **1**. The most recent month, Aug-2018 (-4.00%), is immediately preceded by a positive July (+0.53%).

Evidence: `consecutive_decline_months` column, computed via a gaps-and-islands streak calculation in both SQL (`sql/04_business_cases/01_revenue_decline_detection.sql`) and Python (`python/04_growth_analysis.ipynb`) — every value in the series is 0 or 1, never 2+.

Business Meaning: A single down month should not, on its own, trigger an "the business is declining" narrative. This directly corrects an earlier unverified claim (from this project's original README draft) describing the ending months as a "2-month consecutive decline" — that claim does not hold up against the actual data and has been dropped from this version.

Recommended Action: Use the streak length (not a single month's figure) as the trigger for a management review — e.g., flag when `consecutive_decline_months >= 2`, which has not yet occurred in this dataset.

5. Recent performance (last 3 vs. prior 3 months) is flat, not deteriorating

Finding: Jun-Aug 2018 averaged 1,022,829/month vs. 1,039,333/month for Mar-May 2018 — a -1.59% change.

Evidence: `sql/04_business_cases/04_recent_performance.sql` and `python/04_growth_analysis.ipynb`.

Business Meaning: A -1.59% change over a 3-month window, sitting within a revenue band the business has occupied since Apr-2018, reads as normal month-to-month noise around a stable plateau, not a trend reversal.

Recommended Action: Continue monitoring on a rolling 3-vs-3-month basis rather than reacting to any single month; revisit if the delta exceeds roughly the historical volatility band (`rolling_3m_std`, which has ranged 14,295-23,116 in this same recent window).